# Hongao ZHU

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## Education

**University of California San Diego (UCSD),** California, The United States 2025.09 – present

*PhD in Computational Linguistics*

Advisor: Alex Warstadt

**Shanghai Jiao Tong University (SJTU),** Shanghai, China 2021.09 – 2025.06

*B.A. in English (Linguistics), Wenzhi New Scholar Awardee, GPA 3.9*

Thesis title: A Comparative Study of Language Acquisition in LLMs and Children: Evidence from the *Ba* Construction in Chinese

Advisor: Hai Hu

## Publications

Yikang Liu, Yeting Shen, **Hongao Zhu**, Lilong Xu, Zhiheng Qian, Siyuan Song, Kejia Zhang, Jialong Tang, Pei Zhang, Baosong Yang, Rui Wang, Hai Hu. (2026). A Systematic Assessment of Language Models with Linguistic Minimal Pairs in Chinese. *Transactions of the Association for Computational Linguistics. 14: 755–771.*

Yi-Chien Lin, **Hongao Zhu**, William Schuler. (2026). Vectors from Larger Language Models Predict Human Reading Time and fMRI Data More Poorly when Dimensionality Expansion is Controlled. To appear in *EMNLP 2026*.

**Hongao Zhu**, Lei Lei. (2025). Detecting Authorship Between Generative Models and Humans: A Burrows's Delta Approach. *Digital Scholarship in the Humanities. 40(3): 1064–1075.*

Byung-Doh Oh, **Hongao Zhu**, William Schuler. (2025). The Inverse Scaling Effect of Pre-Trained Language Model Surprisal Is Not Due to Data Leakage. *Findings of the Association for Computational Linguistics: ACL 2025*.

## Preprints

**Hongao Zhu**, Muxiaoqiao Xu, Yikang Liu, Siyuan Song, Yuxia Wang, Byung-Doh Oh, Hai Hu. (2026). A Systematic Analysis of the Predictive Power of LM Surprisal in Reading Chinese. *ArXiv preprint*.

Siyuan Song, Zhiheng Qian, Yunhao Zhang, Linyang He, Xiaozhe Ji, Yingxin Lin, **Hongao Zhu**, Chongtian Shao, Chuhan Lang, Luan Li, Rui Wang, Renfen Hu, Shaonan Wang, Hai Hu. (2026). The First ChineseBabyLM Challenge: Training Data-Efficient and Cognitively Plausible Language Models for Chinese. *ArXiv preprint*.

Qingwen Zhao*, **Hongao Zhu***, Yunqi He, Rui Wang, Aijun Huang, Hai Hu. (2026). LLMs for Automatic Annotation of Mandarin Narrative Transcripts. *ArXiv preprint*. *Equal contribution.

## Manuscripts

**Hongao Zhu**, Alex Warstadt. (2026). Manipulating Plasticity Decay Shows Critical Period Effects for L2 Acquisition in Neural Language Models. *Unpublished manuscript*.

**Hongao Zhu**, Hai Hu. (in prep). A Comparative Study of Language Acquisition in LLMs and Children: Evidence from the *Ba* Construction in Chinese.

**Hongao Zhu**, Dhanashree Kulkarni, Maggie Zhang. (2026). Morphological Complexity vs. Frequency: What Dominates the Learnability of Voice? *Unpublished manuscript*.

**Hongao Zhu**. (2026). Testing the Learnability of Māori Passives. *Unpublished manuscript*.

**Hongao Zhu**. (2026). Quantifying the Information Distribution over Lexical Tones: The Case of Mandarin Chinese. *Unpublished manuscript*.

**Hongao Zhu**. (2025). IPA Illustration of Chongqing Chinese. *Unpublished manuscript*.

## Research Experience

**Critical Period Effects in Neural Language Models** 2025.10 – Present

***PhD Project,*** *supervised by* ***Alex Warstadt*** *Department of Linguistics, UCSD*

- Simulated a critical period in GPT-2 by gradually freezing the neurons with the highest Integrated Gradients attribution during L1→L2 training.
- Trained English, Mandarin, and Dutch models, varying L1 exposure, the L1→L2 switch, and the freezing ceiling, decay rate, and onset.
- Evaluated with BLiMP-style minimal pairs (linguistics), perplexity (computer science), and Δ log-likelihood (cognitive science).
- Found that freezing produced critical-period effects mainly when L1 input stopped abruptly, and that L2 exposure predicted L2 attainment better than plasticity.

**The First ChineseBabyLM Challenge** 2026.04 – 2026.08

***Co-organizer*** *NLPCC 2026 Shared Task*

- Hosted 18 teams who trained Chinese LMs from scratch on at most 102M words and submitted 28 models across three evaluation tracks.
- Helped verify the code for a GPT-BERT baseline, which mixes causal and masked language modeling.

**LLMs for Automatic Annotation of Mandarin Narrative Transcripts** 2025.03 – 2026.08

***Research Assistant,*** *supervised by* ***Aijun Huang*** *and* ***Hai Hu*** *Department of Linguistics, SJTU*

- Built a pipeline that prompts API and local LLMs to annotate narrative macrostructure in spoken Mandarin and measures agreement with human annotators.
- Compared four LLMs on child, young adult, and older adult narratives; the best agreed with humans at κ = .794 (human–human: .872) and cut annotation time by 65%.

**Assessing the Comprehension and Production in Transformer-based Large Language Models** 2024.07 – Present

***Undergraduate Thesis,*** *supervised by* ***Hai Hu*** *Department of Linguistics, SJTU*

- Designed *Ba*-BLiMP, the first constraint-driven benchmark of the Chinese *Ba* construction and validated it using eleven LLMs.
- Trained four LMs from 14M to 410M on a carefully curated Chinese dataset featuring about 3.5B tokens and recorded their acquisition trajectories.
- Transferred methods from First Language Acquisition to obtain open-ended generations from LMs and subsequently conducted error analysis.

**Language Models As Surprisal Predictors to Human Reading Times** 2024.03 – Present

***Research Assistant,*** *supervised by* ***William Schuler*** *Department of Linguistics, Ohio State University*

- Contributed to the psycholinguistic toolkit modelblocks by adding modules for deduplication and trimming human-response data.
- Deduplicated the NaturalStories corpus of Pythia training data, consolidating the predictive power of small LMs proposed by Oh and Schuler (2023).
- Reassessed transformer-based models' alignment with humans on cognitive tasks and behaviors, following Schrimpf et al. (2021).

**Detecting Authorship Between Generative AI Models and Humans: A Burrows's Delta Approach** 2023.10 – 2025.06

***Team Leader,*** *supervised by* ***Lei Lei*** *Department of Linguistics, Shanghai International Studies University*

- Appropriated the classic stylometric algorithm Burrows's Delta into AI detection and proved its robustness on the ArguGPT Corpus.
- Implemented and tested the Delta algorithm, SVM, and RNN using Python and with NLP packages such as Scikit-Learn, SciPy, and NumPy.
- Overrode the classic Delta implementations in Scikit-Learn for efficiency according to the peculiar structures of data.

**ZhoBLiMP: a Systematic Assessment of Language Models with Linguistic Minimal Pairs in Chinese** 2023.10 – 2026.05

***Research Assistant,*** *supervised by* ***Hai Hu*** *Department of Linguistics, SJTU*

- Designed and implemented ZhoBLiMP, the largest Chinese BLiMP containing 118 paradigms covering 15 linguistic phenomena.
- Trained twenty LMs of different sizes (14M–1.4B) on up to three billion Chinese tokens and evaluated them with fourteen off-the-shelf LLMs.
- Modulated ZhoBLiMP for easy scale-ups, by presenting it with a toolchain that allows customizable grammar and vocabulary.

# Invited Talks

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**What can LM Tell Us about Child Language Acquisition: The Case of *Ba* in Mandarin Chinese** 2026.01

Meaning and Language Development Lab, UCSD.

**What can LM Tell Us about Child Language Acquisition: The Case of *Ba* in Mandarin Chinese** 2026.01

Learning, Meaning, and Natural Language Lab, UCSD.

**What can LM Tell Us about Child Language Acquisition: The Case of *Ba* in Mandarin Chinese** 2025.02

Linguistics Brownbag, SJTU.

## Teaching

- **Teaching Assistant, Deep Learning for Natural Language Understanding.** UCSD, Professor Alex Warstadt. 2026.09 – present
- **Teaching Assistant, The Psycholinguistics of Gesture and Sign Languages.** UCSD, Professor Rachel Mayberry. 2026.03 – 2026.06
- **Teaching Assistant, Sign Languages and Their Cultures.** UCSD, Professor Rachel Mayberry. 2026.01 – 2026.03

## Service

- **Organizer, Linguistics BrownBag.** Department of Linguistics, UCSD. 2026.09 – present
- **Lab Meeting Organizer, LeMN Lab.** UCSD, Professor Alex Warstadt. 2026.09 – present
- **Co-organizer, The First ChineseBabyLM Challenge.** NLPCC 2026. 2026.04 – 2026.08
- **Reviewer, Association for Computational Linguistics Rolling Review (ARR).** 2025.01 – present
- **Reviewer, Digital Scholarship in the Humanities.** 2024.10 – present

## Awards and Honors

- Wenzhi New Scholar Award (“文治新人”) 2026
- Second Runner-up and First Prize, 2022 China International Law of the Sea Moot Court Competition 2022
- Champion, RoboCup Small Group in Shanghai Jiao Tong University 2021
- Qualified, Test for English Majors – Band 8 (TEM-8) 2025
- Level 3, China Accreditation Test for Translators and Interpreters (CATTI) 2023
- Level 3, National Computer Rank Examination (NCRE) 2022

## Course Projects

**Learnability of English Voice,** DSC 291, UCSD, California, The United States 2026.06

- Trained LMs on 1M–100M English words with the share of passives set from 0 to 100%; preference for passives grew with that share at a similar rate across corpus sizes.

**Learnability of Māori Passives,** LIGN 220, UCSD, California, The United States 2026.03

- Trained four GPT-NeoX models on Māori text with active or passive forms of 21 verbs removed; the models generalized more easily from passives to actives, supporting the passive as the base form.

**Information Distribution over Mandarin Tones,** LIGN 211, UCSD, California, The United States 2026.03

- Predicted Mandarin tones from segmental features, F0 contours, or both on 1.1M vowels from AISHELL-1; F0 carried most of the information, and vowel height and rounding added a little more.

**IPA Illustration of Chongqing Chinese,** LIGN 210, UCSD, California, The United States 2025.12

- Described the sound system of Chongqing Chinese from a native speaker’s recordings, with acoustic measurements of consonants, vowels, and tones and a glossed North Wind and the Sun.

**RL Logistics Simulator,** SJTU, Shanghai, China 2024.05

- Designed a smart logistics simulator using DQN and Dijkstra that can prevent 10% of the traffic jams in Black Friday events via prediction.

**Acquisition of English Plural Marking by a Cantonese-English Bilingual Child: A Corpus-based study**, SJTU, Shanghai, China 2023.09

- Presented a CHILDES-based study of simultaneous bilingual children to support the interdependent development hypothesis.

## Skills

**Languages:** Mandarin Chinese (Native), English (Advanced), French (Intermediate), Latin (Preliminary)

**Programming Languages:** Python (Scikit-learn, PyTorch, NLTK, Transformers, Pandas, Numpy…), Bash, R, LaTeX, Makefile, C/C++, C#

**Software:** Kubernetes on Nautilus (NRP), ~700 GPU jobs (captain-nemo); Captum (Integrated Gradients); mixed-effects regression